

TRIVIALIZATION

Introduction

A continuous map $f: S \rightarrow A$ between topological spaces is often thought of as describing a “continuous” family of sets $(f^{-1}(a))_{a \in A}$ parametrized by the space A . From this viewpoint the *trivial* maps are the simplest. One says that f is **trivial** if f looks like a projection map $A \times F \rightarrow A$, precisely, if there are a topological space F and a homeomorphism $h: S \rightarrow A \times F$ such that the following diagram commutes.

$$\begin{array}{ccc} S & \xrightarrow[h]{\sim} & A \times F \\ & \searrow f & \swarrow \\ & A & \end{array}$$

Note that then all fibers $f^{-1}(a)$ are homeomorphic to F , and that $h = (f, \lambda)$ for some unique continuous map $\lambda: S \rightarrow F$. We also say that f is trivial over a subspace B of A if the restriction $f|_{f^{-1}(B)}: f^{-1}(B) \rightarrow B$ is trivial. (Then all fibers $f^{-1}(b)$ with $b \in B$ are mutually homeomorphic.)

As in the previous chapter we fix an *o-minimal expansion* $\mathcal{R} = (R, <, S)$ of an ordered (necessarily real closed) field $(R, <, 0, 1, +, -, \cdot)$.

The main result of the first section of this chapter generalizes a well-known theorem of Hardt [29] on semialgebraic continuous maps between real semialgebraic sets, and is roughly that if f , S , and A as above are definable—in the sense of $\mathcal{R}$ of course—then one can partition A into finitely many definable subsets over each of which f is trivial. This is even true in the stronger sense that the corresponding F ’s and h ’s can also be taken to be definable. This result was already established in Chapter 8, (2.14), exercise 4, but there it depended on a model-theoretic argument. The constructive proof given here is more elementary, though much longer. We also give some further refinements, which in Section 2 lead to applications like the conical local structure of definable sets. In Section 3 we show how Wilkie’s theorem used in combination with our trivialization results implies new topological finiteness properties of polynomials with few terms. We solve in particular a problem of Risler and Benedetti on semialgebraic sets of bounded additive complexity. In this way we continue a theme initiated by Khovanskii [32, 33].

§1. Trivialization theorem

(1.1) Consider a definable map $f: S \rightarrow A$, where $A \subseteq \mathbb{R}^m$ and $S \subseteq \mathbb{R}^n$ are definable sets. Viewing A as a **base space** or parameter space, f describes the **family** of sets $(f^{-1}(a))_{a \in A}$. A **definable trivialization** of f is a pair (F, λ) consisting of a definable set $F \subseteq \mathbb{R}^N$, for some N , and a definable map $\lambda: S \rightarrow F$ such that $(f, \lambda): S \rightarrow A \times F$ is a homeomorphism.

So (f, λ) identifies S with the cartesian product $A \times F$, and under this identification f corresponds to the projection map $A \times F \rightarrow A$. Note that then f and λ are continuous, and that (f, λ) maps each fiber $f^{-1}(a)$ of f homeomorphically onto $\{a\} \times F$, in particular all fibers are definably homeomorphic to F .

We call f **definably trivial** if f has a **definable trivialization**. Given a definable subset $A' \subseteq A$ we call f **definably trivial over A'** if the restriction $f|_{f^{-1}(A')}: f^{-1}(A') \rightarrow A'$ is definably trivial. Note that if f is definably trivial, then f is definably trivial over each definable subset of the base space A : if (F, λ) is a definable trivialization of f and $A' \subseteq A$ is definable, then $(F, \lambda|_{f^{-1}(A')})$ is a definable trivialization of $f|_{f^{-1}(A')}: f^{-1}(A') \rightarrow A'$.

We can now formulate the main theorem of this section.

(1.2) **THEOREM.** *Let $f: S \rightarrow A$ be a continuous definable map as above. Then there is a finite partition $A = A_1 \cup \dots \cup A_M$ of the base space A into definable sets A_i such that f is definably trivial over each A_i .*

We actually provide a stronger version where the definable trivializations over each A_i are also compatible with distinguished definable subsets of S .

First we establish an easy lemma on good directions. Next we prove a technical lemma involving homotopies and retractions, and then we can trivialize.

(1.3) Let $n > 0$, and $S^{n-1} := \{x \in \mathbb{R}^n : \|x\| = 1\}$, the set of unit vectors. Recall that a unit vector $u \in S^{n-1}$ is a **good direction** for a set $T \subseteq \mathbb{R}^n$ if each line with direction u intersects T in only finitely many points. (A line with direction u is a set $p + Ru$ with $p \in \mathbb{R}^n$.) Otherwise u is a **bad direction** for T .

If T is definable and $\dim(T) < n$ then it follows from the good directions lemma (4.2) in Chapter 7 that almost all unit vectors $u \in S^{n-1}$ are good directions for T , in the sense that the set of bad directions is of dimension $< n - 1$.

(1.4) **LEMMA.** *Let $T \subseteq \mathbb{R}^{m+n}$ be definable, $\dim(T) < m + n$, $n > 0$. Then there exists a unit vector $u \in S^{n-1}$ such that the set*

$$\{a \in \mathbb{R}^m : u \text{ is a bad direction for } T_a\}$$

has dimension $< m$, that is, u is a good direction for almost all sets T_a .

PROOF. Note that $\dim\{a \in R^m : \dim(T_a) = n\} < m$, so we may as well discard from T the subsets $\{a\} \times T_a$ for which $\dim(T_a) = n$. Hence we have reduced to the case that $\dim(T_a) < n$ for all $a \in R^m$. For each $a \in R^m$ let $B_a \subseteq S^{n-1}$ be the set of bad directions for T_a . Then $\dim(B_a) < n-1$ for all $a \in R^m$, so $\dim(B) < m+n-1$ where $B := \bigcup\{\{a\} \times B_a : a \in R^m\} \subseteq R^m \times S^{n-1}$. Let $p: R^m \times S^{n-1} \rightarrow S^{n-1}$ be the projection map onto the second factor. Then it follows from Chapter 4, Section 1 that there is $u \in S^{n-1}$ such that $\dim(p^{-1}(u) \cap B) < m$. This vector u is a good direction for almost all T_a . $\square$

The following technical lemma is the key to the trivialization theorem. To formulate it we need a mild generalization of cells: define a **generalized open cell in R^m** to be a definable open subset of R^m that is definably homeomorphic to R^m . Examples are open cells in R^m and m -simplexes in R^m . Clearly each generalized open cell is definably contractible to each of its points.

(1.5) LEMMA. *Let $S_1, \dots, S_k \subseteq R^{m+n}$ be definable, let $A \subseteq R^m$ be definable, and let $\pi: R^{m+n} \rightarrow R^m$ be the projection map. Then there are disjoint generalized open cells $A_1, \dots, A_h \subseteq A$ such that $\dim(A - \bigcup A_i) < m$, and such that for each $i \in \{1, \dots, h\}$ and each definable contraction $H: A_i \times [0, 1] \rightarrow A_i$ from A_i to a point $a \in A_i$ there exist a definable retraction $r: A_i \times R^n \rightarrow \{a\} \times R^n$ and a definable contraction $\tilde{H}: A_i \times R^n \times [0, 1] \rightarrow A_i \times R^n$ from $A_i \times R^n$ to r , with the following properties:*

- (1) $\tilde{H}$ lifts H , that is, the following diagram commutes:

$$\begin{array}{ccc} A_i \times R^n \times [0, 1] & \xrightarrow{\tilde{H}} & A_i \times R^n \\ \pi_i \times \text{id} \downarrow & & \downarrow \pi_i \\ A_i \times [0, 1] & \xrightarrow{H} & A_i \end{array}$$

where $\pi_i = \pi|_{A_i \times R^n}$.

- (2) For each S_j the maps r and $\tilde{H}$ yield by restriction a definable retraction

$$r_j: (A_i \times R^n) \cap S_j \rightarrow (\{a\} \times R^n) \cap S_j$$

and a definable contraction

$$\tilde{H}_j: ((A_i \times R^n) \cap S_j) \times [0, 1] \rightarrow (A_i \times R^n) \cap S_j$$

from $(A_i \times R^n) \cap S_j$ to r_j .

- (3) For each $x \in A_i$ the map r restricts to a definable homeomorphism

$$r_x: \{x\} \times R^n \rightarrow \{a\} \times R^n$$

that maps $(\{x\} \times R^n) \cap S_j$ onto $(\{a\} \times R^n) \cap S_j$ for each j .

PROOF. By induction on n . The case $n = 0$ follows by cell decomposition. In more detail: take a finite partition of A into cells that is compatible with $S_1, \dots, S_k$, and let $A_1, \dots, A_h$ be the open cells in this partition; then H and the point $a \in A_i$ determine r , and by taking $\tilde{H} = H$ the conditions (1), (2), and (3) are trivially satisfied.

Let $n > 0$ and assume inductively that the assertion holds for $n - 1$ instead of n . We then first do the case that $S_1, \dots, S_k$ are all *bounded*. Let

$$T := \bigcup_j \text{bd}(S_j),$$

so that T is closed and bounded, and $\dim(T) < m + n$. (Later we modify T to a set $\tilde{T}$ that is the graph of a full multivalued function.) By the previous lemma there is a unit vector $u \in S^{n-1}$ that is a good direction for almost all sets T_a , that is, $\dim(B) < m$ where $B := \{a \in R^m : u \text{ is a bad direction for } T_a\}$.

Recall that $e_1, \dots, e_n$ denotes the standard basis of the vector space R^n over R . We view e_n as pointing in the “vertical direction”. Take a linear automorphism $\lambda: R^n \rightarrow R^n$ such that $\lambda(u) = e_n$, and replace each set S_j by its image $\{(x, \lambda(y)) : (x, y) \in S_j\}$ under the automorphism $\text{id} \times \lambda$ of R^{m+n} . This is permitted since a solution to our problem for the new S_j ’s can be transformed back via $\text{id} \times \lambda^{-1}$ to a solution of the problem for the original S_j ’s. Note also that then T gets replaced by $\{(x, \lambda(y)) : (x, y) \in T\}$, so our new T satisfies

$$\dim(B) < m, \text{ where } B = \{a \in R^m : e_n \text{ is a bad direction for } T_a\}.$$

Now let $p: R^{m+n} \rightarrow R^{m+n-1}$ be the projection map on the first $m+n-1$ coordinates and $q: R^{m+n-1} \rightarrow R^m$ the projection map on the first m coordinates, so that $\pi = q \circ p$. Then e_{m+n} is a good direction for $T \cap \pi^{-1}(A - B)$. Hence by cell decomposition the set $T \cap \pi^{-1}(A - B)$ is the disjoint union of $\Gamma(f)$ ’s for finitely many definable continuous functions f defined on cells C that form a finite partition $\mathcal{P}$ of $(A - B) \times R^{n-1}$. Call these functions on cells $C \in \mathcal{P}$ **distinguished**. Take a triangulation (Φ, K) of $(A - B) \times R^{n-1}$ compatible with each cell $C \in \mathcal{P}$. Let F be the collection of the restrictions $f|_D$ of the distinguished functions f on cells $C \in \mathcal{P}$ to sets $D \in \Phi^{-1}(K)$ with $D \subseteq C$. Then F is a multivalued function on $((A - B) \times R^{n-1}, \Phi^{-1}(K))$ in the sense of (2.5) of Chapter 8, and F is closed by (2.6) of Chapter 8, since $\Gamma(F) = T \cap \pi^{-1}(A - B)$ is closed in $\pi^{-1}(A - B) = (A - B) \times R^n$. Unfortunately, F may not be full, and therefore we modify F as follows to a full multivalued function $\tilde{F}$. First extend each $f \in F|_D$ continuously to the closure $\text{cl}(D) \cap (A - B)$ of D in $A - B$, and then further to a bounded continuous definable function $\tilde{f}: A - B \rightarrow R$, using (3.10) of Chapter 8. Now put

$$\tilde{T} := \left(\bigcup_{f \in F} \Gamma(\tilde{f}) \right) \cup \Gamma(f_0),$$

where $f_0: (A - B) \times R^{n-1} \rightarrow R$ is the function that is identically 0.

Next replace (Φ, K) by a new triangulation $(\tilde{\Phi}, \tilde{K})$ of $(A - B) \times R^{n-1}$ that is compatible with all sets in $\Phi^{-1}(K)$, all sets $p(S_j \cap \Gamma(\tilde{f}))$ for $f \in F$, and all sets $p(S_j \cap \Gamma(f_0))$. In addition we arrange that the nonempty restrictions of the $\tilde{f}$'s for $f \in F$ and of f_0 to the sets in $\tilde{\Phi}^{-1}(\tilde{K})$ form a multivalued function $\tilde{F}$ on the triangulated set $((A - B) \times R^{n-1}, \tilde{\Phi}^{-1}(\tilde{K}))$.

Note that $\Gamma(\tilde{F}) = \tilde{T}$, that $\tilde{T}$ is closed in $(A - B) \times R^n$, and that $T \cap ((A - B) \times R^n) \subseteq \tilde{T}$. Clearly $\tilde{F}$ is full. Let $\tilde{F}|D = \{f_{D1}, \dots, f_{Dk(D)}\}$ with $f_{D1} < \dots < f_{Dk(D)}$, $k(D) \geq 1$, where $D \in \tilde{\Phi}^{-1}(\tilde{K})$. The compatibility of $\tilde{\Phi}^{-1}(\tilde{K})$ with the sets indicated, together with (3.3) of Chapter 6, implies that for any $f, g \in \tilde{F}|D$ and set S_j we have:

$$(*) \quad \begin{cases} \text{either } \Gamma(f) \subseteq S_j, \text{ or } \Gamma(f) \cap S_j = \emptyset; \\ \text{if } f < g \text{ are successive members of } \tilde{F}|D, \\ \text{then either } (f, g) \subseteq S_j, \text{ or } (f, g) \cap S_j = \emptyset; \\ \text{if } f = f_{D1} \text{ is the first member of } \tilde{F}|D, \text{ then } (-\infty, f) \cap S_j = \emptyset; \\ \text{if } g = f_{Dk(D)} \text{ is the last member of } \tilde{F}|D, \text{ then } (g, +\infty) \cap S_j = \emptyset. \end{cases}$$

Now apply the inductive hypothesis to the sets in $\tilde{\Phi}^{-1}(\tilde{K})$ and $A - B$ instead of A . This gives disjoint generalized open cells $A_1, \dots, A_h \subseteq A - B$ with $\dim(A - \bigcup A_i) < m$ and with the following properties for any given $i \in \{1, \dots, h\}$:

- (1)' Each definable contraction $H: A_i \times [0, 1] \rightarrow A_i$ from A_i to a point $a \in A_i$ can be lifted to a definable contraction $H': A_i \times R^{n-1} \times [0, 1] \rightarrow A_i \times R^{n-1}$ from $A_i \times R^{n-1}$ to a definable retraction $r': A_i \times R^{n-1} \rightarrow \{a\} \times R^{n-1}$.
- (2)' For each $D \in \tilde{\Phi}^{-1}(\tilde{K})$ the maps r' and H' from (1)' yield by restriction a definable retraction $r'_D: (A_i \times R^{n-1}) \cap D \rightarrow (\{a\} \times R^{n-1}) \cap D$ and a definable contraction $H'_D: [(A_i \times R^{n-1}) \cap D] \times [0, 1] \rightarrow (A_i \times R^{n-1}) \cap D$ from $(A_i \times R^{n-1}) \cap D$ to r'_D .
- (3)' For each $x \in A_i$ the map r' from (1)' restricts to a definable homeomorphism $r'_x: \{x\} \times R^{n-1} \rightarrow \{a\} \times R^{n-1}$ mapping $(\{x\} \times R^{n-1}) \cap D$ onto $(\{a\} \times R^{n-1}) \cap D$, for each $D \in \tilde{\Phi}^{-1}(\tilde{K})$.

Fix an A_i and write $A_i \times R^n$ as the disjoint union of the following sets:

$$\begin{aligned} G_{Dk} &:= \{(y, f_{Dk}(y)) : y \in (A_i \times R^{n-1}) \cap D\}, \quad D \in \tilde{\Phi}^{-1}(\tilde{K}), \quad 1 \leq k \leq k(D); \\ U_{Dk} &:= \{(y, s) : y \in (A_i \times R^{n-1}) \cap D, \quad f_{Dk}(y) < s < f_{Dk+1}(y)\}, \\ &\quad D \in \tilde{\Phi}^{-1}(\tilde{K}), \quad 1 \leq k < k(D); \\ U_{D-} &:= \{(y, s) : y \in (A_i \times R^{n-1}) \cap D, \quad s < f_{D1}(y)\}, \quad D \in \tilde{\Phi}^{-1}(\tilde{K}); \\ U_{D+} &:= \{(y, s) : y \in (A_i \times R^{n-1}) \cap D, \quad s > f_{Dk(D)}(y)\}, \quad D \in \tilde{\Phi}^{-1}(\tilde{K}). \end{aligned}$$

By (*) every set $(A_i \times R^n) \cap S_j$ is a union of some of these sets.

Let a definable contraction $H: A_i \times [0, 1] \rightarrow A_i$ from A_i to a point $a \in A_i$ be given, and let H' and r' be as in (1)', (2)' and (3)' above. We now show how to lift H' to

a definable homotopy $\tilde{H}: A_i \times R^n \times [0, 1] \rightarrow A_i \times R^n$ by defining $\tilde{H}$ on each of the sets $G_{Dk} \times [0, 1]$, $U_{Dk} \times [0, 1]$, $U_{D-} \times [0, 1]$ and $U_{D+} \times [0, 1]$:

- (i) $\tilde{H}(y, f_{Dk}(y), t) := (H'(y, t), f_{Dk}(H'(y, t)))$, for $(y, f_{Dk}(y)) \in G_{Dk}$, $t \in [0, 1]$;
- (ii) write each $z \in U_{Dk}$ uniquely as $z = (y, (1 - u)f_{Dk}(y) + uf_{Dk+1}(y))$ with $0 < u < 1$ and set, for $t \in [0, 1]$,

$$\tilde{H}(z, t) := (H'(y, t), (1 - u)f_{Dk}(H'(y, t)) + uf_{Dk+1}(H'(y, t)));$$

- (iii) write each $z \in U_{D-}$ uniquely as $z = (y, f_{D1}(y) - u)$ with $u > 0$ and set $\tilde{H}(z, t) := (H'(y, t), f_{D1}(H'(y, t)) - u)$, for $t \in [0, 1]$;
- (iv) write each $z \in U_{D+}$ uniquely as $z = (y, f_{Dk(D)}(y) + u)$ with $u > 0$ and set $\tilde{H}(z, t) := (H'(y, t), f_{Dk(D)}(H'(y, t)) + u)$, for $t \in [0, 1]$.

Clearly $\tilde{H}$ maps $G_{Dk} \times [0, 1]$ into G_{Dk} , $U_{Dk} \times [0, 1]$ into U_{Dk} , $U_{D-} \times [0, 1]$ into U_{D-} , and $U_{D+} \times [0, 1]$ into U_{D+} . A lengthy but routine argument using fullness of $\tilde{F}$ (and for instance lemma (4.2) from Chapter 6) shows that $\tilde{H}$ is continuous. Hence $\tilde{H}$ is a definable contraction from $A_i \times R^n$ to the retraction $r: A_i \times R^n \rightarrow \{a\} \times R^n$ given by $r(z) := \tilde{H}(z, 1)$. One checks easily that $\tilde{H}$ and r have the properties required in (1) and (2). To check (3) one shows first that for fixed $y \in A_i \times R^{n-1}$ we have $r(y, s) = (r'(y), g_y(s))$, where $g_y: R \rightarrow R$ is a strictly increasing bijection. Thus each map $r_x: \{x\} \times R^n \rightarrow \{a\} \times R^n$ as in (3) is bijective. To get the continuity of the inverse of r_x one first shows there is a constant $M > 0$ in R , independent of $y \in A_i \times R^{n-1}$, such that $|g_y(s) - s| \leq M$ for all s , and then one applies lemma (4.2) from Chapter 6, and the inductive assumption (3)' on r'_x .

This settles the case that $S_1, \dots, S_k$ are bounded. Now we reduce the general case to this special case. Let I be the interval $(-1, 1)$ and $\mu: R \rightarrow I$ the homeomorphism given by $\mu(x) := x/\sqrt{1+x^2}$, and let $\mu_p: R^p \rightarrow I^p$ be the homeomorphism given by $\mu_p(x_1, \dots, x_p) := (\mu(x_1), \dots, \mu(x_p))$. Note that μ_p is definable.

Let definable $S_1, \dots, S_k \subseteq R^{m+n}$ and definable $A \subseteq R^m$ be given. Then the sets $S_{\mu i} := \mu_{m+n}(S_i) \subseteq R^{m+n}$ are bounded, and we add to $S_{\mu 1}, \dots, S_{\mu k}$ one extra set, namely $S_{\mu 0} := I^{m+n}$, so $S_{\mu 0} = \mu_{m+n}(R^{m+n})$. Put $A_\mu := \mu_m(A)$. We now apply the result proved for the bounded case to the sets $S_{\mu 0}, \dots, S_{\mu k}$, with A_μ playing the role of A . This gives us disjoint generalized open cells $A_{\mu 1}, \dots, A_{\mu h} \subseteq A_\mu$ with $\dim(A_\mu - \bigcup A_{\mu i}) < m$ such that for any given $i \in \{1, \dots, h\}$ we have:

- (1 $_\mu$) each definable contraction $H_\mu: A_{\mu i} \times [0, 1] \rightarrow A_{\mu i}$ to a point $a_\mu \in A_{\mu i}$ lifts to a definable contraction $\tilde{H}_\mu: A_{\mu i} \times R^n \times [0, 1] \rightarrow A_{\mu i} \times R^n$ from $A_{\mu i} \times R^n$ to a definable retraction $r_\mu: A_{\mu i} \times R^n \rightarrow \{a_\mu\} \times R^n$;
- (2 $_\mu$) for each $S_{\mu j}$ (including $S_{\mu 0}$) the maps r_μ and $\tilde{H}_\mu$ from (1 $_\mu$) yield by restriction a definable retraction $r_{\mu j}: (A_{\mu i} \times R^n) \cap S_{\mu j} \rightarrow (\{a_\mu\} \times R^n) \cap S_{\mu j}$, and a definable contraction $\tilde{H}_{\mu j}: ((A_{\mu i} \times R^n) \cap S_{\mu j}) \times [0, 1] \rightarrow (A_{\mu i} \times R^n) \cap S_{\mu j}$ from $(A_{\mu i} \times R^n) \cap S_{\mu j}$ to $r_{\mu j}$;
- (3 $_\mu$) for each $x \in A_{\mu i}$ the map r_μ from (1 $_\mu$) restricts to a homeomorphism $r_{\mu x}: \{x\} \times R^n \rightarrow \{a_\mu\} \times R^n$ that maps $(\{x\} \times R^n) \cap S_{\mu j}$ onto $(\{a_\mu\} \times R^n) \cap S_{\mu j}$ for each j .

Now we use $\mu_m^{-1} : I^m \rightarrow R^m$ and $\mu_{m+n}^{-1} : I^{m+n} \rightarrow R^{m+n}$ to transform back. Put $A_i := \mu_m^{-1}(A_{\mu i})$, $i = 1, \dots, h$. Clearly $A_1, \dots, A_h$ are generalized open cells and $\dim(A - \bigcup A_i) < m$. Let $H : A_i \times [0, 1] \rightarrow \{a\}$ be a definable contraction from A_i to a point $a \in A_i$. We shall find r and $\tilde{H}$ with properties (1), (2), and (3) of the lemma. Set $a_\mu := \mu_m(a)$ and define $H_\mu : A_{\mu i} \times [0, 1] \rightarrow A_{\mu i}$ by $H_\mu(x, t) := \mu_m H(\mu_m^{-1}(x), t)$. Then H_μ is a definable contraction from $A_{\mu i}$ to a_μ . Let $\tilde{H}_\mu$ and r_μ be as in (1_μ) , (2_μ) , and (3_μ) above. Applying (2_μ) to $S_{\mu 0} = I^{m+n}$ we see that $\tilde{H}_\mu$ maps $A_{\mu i} \times I^n \times [0, 1]$ into $A_{\mu i} \times I^n$, so we may define $\tilde{H} : A_i \times R^n \times [0, 1] \rightarrow A_i \times R^n$ by

$$\tilde{H}(x, y, t) := \mu_{m+n}^{-1} \tilde{H}_\mu(\mu_m(x), \mu_n(y), t).$$

Similarly r_μ maps $A_{\mu i} \times I^n$ into $\{a_\mu\} \times I^n$, by $(2'_\mu)$ applied to $S_{\mu 0}$, so we may define $r : A_i \times R^n \rightarrow \{a\} \times R^n$ by $r(x, y) := \mu_{m+n}^{-1} r_\mu(\mu_m(x), \mu_n(y))$. One verifies easily that then $\tilde{H}$ and r satisfy the statements of the lemma. $\square$

(1.6) In our trivialization theorem we also want to take into account distinguished definable subsets of S . Here are the relevant definitions.

Let $A \subseteq R^m$ and $S \subseteq R^n$ be definable sets, $f : S \rightarrow A$ a continuous definable map, and $S_1, \dots, S_k$ definable subsets of S .

Then a **definable trivialization of f respecting $S_1, \dots, S_k$** consists of a definable trivialization (F, λ) of f and distinguished definable subsets $F_1, \dots, F_k$ of F , such that $(f, \lambda)(S_i) = A \times F_i$ for $i = 1, \dots, k$. Note that then each restriction $f|_{S_i} : S_i \rightarrow A$ has definable trivialization $(F_i, \lambda|_{S_i})$. Given definable $A' \subseteq A$ we call $(f; S_1, \dots, S_k)$ **definably trivial over A'** if the restriction

$$f|_{f^{-1}(A')} : f^{-1}(A') \rightarrow A'$$

has a definable trivialization respecting $S_1 \cap f^{-1}(A'), \dots, S_k \cap f^{-1}(A')$. Note that then $(f; S_1, \dots, S_k)$ is also definably trivial over each definable subset of A' . Also, if $a \in A'$ we can always take a definable trivialization over A' respecting $S_1 \cap f^{-1}(A'), \dots, S_k \cap f^{-1}(A')$ to be of the form $(f^{-1}(a), \lambda)$ with λ the identity on $f^{-1}(a) \subseteq f^{-1}(A')$, and with distinguished subsets $S_1 \cap f^{-1}(a), \dots, S_k \cap f^{-1}(a)$ of $f^{-1}(a)$.

(1.7) TRIVIALIZATION THEOREM. *With $f : S \rightarrow A$ and $S_1, \dots, S_k$ as above, we can partition A into definable subsets $A_1, \dots, A_M$, such that $(f; S_1, \dots, S_k)$ is definably trivial over each A_i .*

PROOF. By induction on $\dim(A)$. If $\dim(A) \leq 0$, then A is finite, and the theorem holds trivially. Let $\dim(A) > 0$, and assume the desired result holds for lower values of $\dim(A)$. By partitioning A into finitely many cells A' , and working with the restrictions $f|_{f^{-1}(A')} : f^{-1}(A') \rightarrow A'$ we may as well assume that A is a cell; then using the homeomorphism p_A from A onto an open cell, we may further assume A is an open cell, say in R^m , so $\dim(A) = m$. Next we replace S by the

reversed graph of f by applying the definable homeomorphism $y \mapsto (f(y), y)$ from S onto this reversed graph. So from now on S is a definable subset of R^{m+n} , and f is the restriction $\pi|_S: S \rightarrow A$ of the projection map $\pi: R^{m+n} \rightarrow R^m$. We need one further reduction to the bounded case. Let $\mu: R \rightarrow I = (-1, 1)$ be the definable homeomorphism given by $\mu(x) = x/\sqrt{1+x^2}$, and let $\mu_p: R^p \rightarrow I^p$ be the induced homeomorphism given by $\mu_p(x_1, \dots, x_p) = (\mu(x_1), \dots, \mu(x_p))$, for $p \in \mathbb{N}$. Put $S_\mu := \mu_{m+n}(S)$, $S_{\mu j} := \mu_{m+n}(S_j)$, $A_\mu := \mu_m(A)$, and let $f_\mu: S_\mu \rightarrow A_\mu$ be the restriction of the projection map $R^{m+n} \rightarrow R^m$. Clearly it suffices to show

(*) $\left\{ \begin{array}{l} A_\mu \text{ can be partitioned into definable subsets } A_{\mu 1}, \dots, A_{\mu M} \text{ such that} \\ (f_\mu; S_{\mu 1}, \dots, S_{\mu k}) \text{ is definably trivial over each } A_{\mu i}. \end{array} \right.$

We distinguish one more subset of R^{m+n} , namely $A_\mu \times \text{cl}(I)^n$, where $I = (-1, 1)$, and apply lemma (1.5), with $S_\mu, S_{\mu 1}, \dots, S_{\mu k}$ and $A_\mu \times \text{cl}(I)^n$ as distinguished subsets of R^{m+n} , and A_μ in the role of A : this gives disjoint generalized open cells $A_{\mu 1}, \dots, A_{\mu h}$ in A_μ with $\dim(A_\mu - \bigcup A_{\mu i}) < m$, and further properties from lemma (1.5). By the inductive assumption we can partition $A_\mu - \bigcup A_{\mu i}$ into finitely many definable subsets over each of which $(f_\mu; S_{\mu 1}, \dots, S_{\mu k})$ is definably trivial. So (*) will follow from

(**) $(f_\mu; S_{\mu 1}, \dots, S_{\mu k})$ is trivial over each $A_{\mu i}$, $1 \leq i \leq h$.

Fix an i and take a definable contraction H from $A_{\mu i}$ to one of its points a . Then lemma (1.5) gives us a definable retraction $r: A_{\mu i} \times R^n \rightarrow \{a\} \times R^n$, which for each $x \in A_{\mu i}$ maps $\{x\} \times R^n$ homeomorphically onto $\{a\} \times R^n$, $\{x\} \times S_{\mu j}$ onto $\{a\} \times S_{\mu j}$, $\{x\} \times S_{\mu j x}$ onto $\{a\} \times S_{\mu j a}$ ($1 \leq j \leq k$), and $\{x\} \times \text{cl}(I)^n$ onto $\{a\} \times \text{cl}(I)^n$.

Write $r(x, y) = (a, s(x, y))$, where $s: A_{\mu i} \times R^n \rightarrow R^n$ is a definable continuous map. Then the map $(x, y) \mapsto (x, s(x, y)): A_{\mu i} \times \text{cl}(I)^n \rightarrow A_{\mu i} \times \text{cl}(I)^n$ is a continuous definable bijection. This bijection must be a homeomorphism: each point in $A_{\mu i}$ has a definable neighborhood N in $A_{\mu i}$ that is closed and bounded, and this bijection maps $N \times \text{cl}(I)^n$ homeomorphically onto itself, by Chapter 6, (1.12). Define $\lambda_i: f_\mu^{-1}(A_{\mu i}) \rightarrow S_{\mu a}$ by $\lambda_i(x, y) := s(x, y)$. It is now easy to verify that $(S_{\mu a}, \lambda_i)$ is a definable trivialization of $(f_\mu; S_{\mu 1}, \dots, S_{\mu k})$ over $A_{\mu i}$. This finishes the proof of (**). $\square$

(1.8) Note that theorem (1.2) is the special case $k = 0$ of theorem (1.7). Readers familiar with elementary model theory will note that the easy model-theoretic arguments (in Chapter 8, (2.14)) proving theorem (1.2) can be used to derive theorem (1.7) in a similar way.

(1.9) The following exercise gives an alternative formulation of the trivialization theorem, and explains why such results are also referred to as “generic” triviality theorems.

EXERCISE. Let $f: S \rightarrow A$ be a continuous definable map between nonempty definable sets S and A . Show that there is a definable set $E \subseteq A$ with $\dim E < \dim A$ such that f is definably trivial over each definably connected component of $A - E$.

§2. Applications

(2.1) An immediate application of the trivialization theorem is that for a given definable set $S \subseteq R^{m+n}$ the definable sets S_x fall into only finitely many different definable homeomorphism types as x ranges over R^m .

To see this, let $f: S \rightarrow R^m$ be the restriction of the projection map $R^{m+n} \rightarrow R^m$, and take a partition of R^m into definable sets $A_1, \dots, A_M$ over each of which f is definably trivial, say with definable trivialization (λ_i, F_i) over A_i . Then S_x is definably homeomorphic to F_i for all $x \in A_i$.

A (well-known) special case of this finiteness result is the fact that for fixed natural numbers d and n the zero sets of polynomials $f(X_1, \dots, X_n) \in \mathbb{R}[X_1, \dots, X_n]$ in $\mathbb{R}^n$ of degree at most d fall into only finitely many semialgebraic homeomorphism types. (Instead of $\mathbb{R}$ we can take here of course any real closed field.)

We can strengthen these results by considering “embedded homeomorphism type”. Two sets $X, Y \subseteq \mathbb{R}^n$ have the same **embedded homeomorphism type** if there is a homeomorphism $h: \mathbb{R}^n \xrightarrow{\sim} \mathbb{R}^n$ such that $h(X) = Y$. To have the same embedded homeomorphism type is an equivalence relation on the collection of subsets of $\mathbb{R}^n$. Similarly we say that two definable sets $X, Y \subseteq R^n$ have the same **embedded definable homeomorphism type** if there is a definable homeomorphism $h: R^n \xrightarrow{\sim} R^n$ such that $h(X) = Y$. To have the same embedded definable homeomorphism type is an equivalence relation on the collection of definable subsets of R^n . Using the “trivialization theorem with distinguished subsets” the same arguments as above show

PROPOSITION. *If $S \subseteq R^{m+n}$ is definable, then the definable sets $S_x \subseteq R^n$ fall into only finitely many embedded definable homeomorphism types, as x ranges over R^m .*

(2.2) Next we give an application to the local structure of definable sets. Given a definable set $A \subseteq R^n$ and a point $p \in R^n$ not in A we define the **cone with base A and vertex p** to be the set $[A, p] := \{ta + (1-t)p : a \in A, 0 \leq t \leq 1\}$, the union of the line segments $[a, p]$ with $a \in A$.

Given $p \in R^n$ and $\epsilon > 0$, we also put

$B(p, \epsilon) := \{x \in R^n : \|x - p\| \leq \epsilon\}$, the closed ball centered at p with radius ϵ ;

$S(p, \epsilon) := \{x \in R^n : \|x - p\| = \epsilon\}$, the sphere centered at p with radius ϵ .

(2.3) **THEOREM.** *Let $E \subseteq R^n$ be a definable set and p a non-isolated point of E . Then there is $\epsilon > 0$ such that $E \cap B(p, \epsilon)$ is definably homeomorphic to the cone with base $E \cap S(p, \epsilon)$ and vertex p . More precisely, there is a definable homeomorphism ϕ from $B(p, \epsilon)$ onto itself, such that*

- (i) $\phi(p) = p$ and ϕ is the identity on $S(p, \epsilon)$,

- (ii) $\|\phi(x) - p\| = \|x - p\|$ for all $x \in B(p, \epsilon)$,
- (iii) $\phi(E \cap B(p, \epsilon)) = [E \cap S(p, \epsilon), p]$.

PROOF. Apply the trivialization theorem to the distance function $x \mapsto \|x - p\| : \mathbb{R}^n \rightarrow [0, \infty)$, with E as a distinguished subset of $\mathbb{R}^n$. Note that for $\epsilon > 0$ the ϵ -fiber of this map is $S(p, \epsilon)$, and that the inverse image of $(0, \epsilon]$ is $B(p, \epsilon) - \{p\}$. This gives a definable trivialization $\lambda : B(p, \epsilon) - \{p\} \rightarrow S(p, \epsilon)$ respecting E over a half-interval $(0, \epsilon]$, $\epsilon > 0$, with $\lambda|_{S(p, \epsilon)} = \text{identity}$.

Hence we have a definable homeomorphism

$$x \mapsto (\|x - p\|, \lambda(x)) : B(p, \epsilon) - \{p\} \rightarrow (0, \epsilon] \times S(p, \epsilon)$$

mapping $E \cap (B(p, \epsilon) - \{p\})$ onto $(0, \epsilon] \times (E \cap S(p, \epsilon))$.

Now define $\phi : B(p, \epsilon) \rightarrow B(p, \epsilon)$ by $\phi(x) = p + (\|x - p\|/\epsilon) \cdot (\lambda(x) - p)$ for $x \neq p$, and $\phi(p) = p$. One verifies easily that this ϕ has all the desired properties. $\square$

§3. On a conjecture of Benedetti and Risler

(3.1) The generic triviality theorem leads to a new result on zero sets of “fewnomials”, as recorded in the following proposition. We will make essential use of the o-minimality of the model-theoretic structure $\mathbb{R}_{\text{exp}} := (\mathbb{R}, <, 0, 1, +, -, \cdot, \exp)$ due to Wilkie [64]; see also [22] for another proof that $\mathbb{R}_{\text{exp}}$ is o-minimal.

(3.2) PROPOSITION. *For any given natural numbers m and n there are only finitely many homeomorphism types among the sets $Z(f) := \{x \in \mathbb{R}^n : f(x) = 0\}$, where $f(X_1, \dots, X_n) \in \mathbb{R}[X_1, \dots, X_n]$ has at most m monomials.*

PROOF. We are of course going to use the fact that the function

$$(x, y) \mapsto x^y := \exp(y \log(x)) : (0, \infty) \times \mathbb{R} \rightarrow \mathbb{R}$$

is definable in $\mathbb{R}_{\text{exp}}$. A minor complication is that, on the other hand, there is no function $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ definable in $\mathbb{R}_{\text{exp}}$ such that $F(x, k) = x^k$ for all $x \in \mathbb{R}$ and $k \in \mathbb{N}$. (See exercise (3.9).) Fortunately we can instead define two functions $E, O : \mathbb{R}^2 \rightarrow \mathbb{R}$ in $\mathbb{R}_{\text{exp}}$ such that $E(x, k) = x^k$ for all $x \in \mathbb{R}$ and $k \in 2\mathbb{N}$ and $O(x, k) = x^k$ for all $x \in \mathbb{R}$ and $k \in 2\mathbb{N} + 1$. We define E and O as follows:

$$E(x, y) := \begin{cases} |x|^y & \text{for } x \neq 0, \\ 1 & \text{for } x = y = 0, \\ 0 & \text{for } x = 0, y \neq 0; \end{cases}$$

$$O(x, y) := \begin{cases} x^y & \text{for } x > 0, \\ -(-x)^y & \text{for } x < 0, \\ 0 & \text{for } x = 0. \end{cases}$$

Let us now fix natural numbers m and n , and let $f = \sum_{i=1}^m a_i X^\alpha$ denote an arbitrary polynomial in $X = (X_1, \dots, X_n)$ over $\mathbb{R}$ with at most m monomials: $a_i \in \mathbb{R}$ and $\alpha_i = (\alpha_{i1}, \dots, \alpha_{in}) \in \mathbb{N}^n$ for $i = 1, \dots, m$. Let

$$\text{Even}_i(f) := \{j : 1 \leq j \leq n, \alpha_{ij} \in 2\mathbb{N}\} \text{ and } \text{Odd}_i(f) := \{1, \dots, n\} - \text{Even}_i(f)$$

for $i = 1, \dots, m$. Then we have for all $x \in \mathbb{R}^n$

$$f(x) = \sum_i a_i \prod_{j \in \text{Even}_i(f)} E(x_j, \alpha_{ij}) \prod_{j \in \text{Odd}_i(f)} O(x_j, \alpha_{ij}).$$

In this expression we may consider the $(m + mn)$ -tuple

$$c(f) := (a_1, \dots, a_m, \alpha_{11}, \dots, \alpha_{1n}, \dots, \alpha_{m1}, \dots, \alpha_{mn}) \in \mathbb{R}^{m+mn}$$

as a parameter. Let $\mathbb{E}_1, \dots, \mathbb{E}_K$, with $K = 2^{mn}$, be the different n -tuples of the form $(\text{Even}_1(f), \dots, \text{Even}_m(f))$ as f varies. The considerations above show that there are sets $S_1, \dots, S_K \subseteq \mathbb{R}^{m+mn} \times \mathbb{R}^n$, definable in $\mathbb{R}_{\text{exp}}$, such that if $(\text{Even}_1(f), \dots, \text{Even}_m(f)) = \mathbb{E}_k$, $1 \leq k \leq K$, then $Z(f) = (S_k)_{c(f)}$. (In other words, the zero sets of the real polynomials in $X = (X_1, \dots, X_n)$ with at most m monomials fall into 2^{mn} different definable families, where “definable” here means “definable in $\mathbb{R}_{\text{exp}}$ ”.) The desired result now follows from the fact that among the subsets of $\mathbb{R}^n$ of the form $(S_k)_c$ ($1 \leq k \leq K, c \in \mathbb{R}^{m+mn}$) there are only finitely many homeomorphism types, by (2.1). $\square$

(3.3) REMARK. The same argument shows that for given m, n there are only finitely many different *embedded* definable homeomorphism types among the sets $Z(f) \subseteq \mathbb{R}^n$ of proposition (3.2), where “definable” refers to definability with parameters in $\mathbb{R}_{\text{exp}}$. (Here we use the proposition stated at the end of (2.1).)

(3.4) From a geometric viewpoint the “number of monomials” is not a good complexity measure for polynomials, for example, it misbehaves under linear changes of variables. Therefore Benedetti and Risler [2] introduced the more geometric notion of “additive complexity”, which also makes sense for rational functions. Let us say that a rational function $f \in \mathbb{R}(X)$, with $X = (X_1, \dots, X_n)$, has **additive complexity** $\leq k$ if there are nonzero rational functions $f_1, \dots, f_k \in \mathbb{R}(X)$ such that

$$f_1 = a_1 X^{u_1} + b_1 X^{v_1}$$

with $a_1, b_1 \in \mathbb{R}$, $u_1 = (u_{11}, \dots, u_{1n}) \in \mathbb{Z}^n$, $v_1 = (v_{11}, \dots, v_{1n}) \in \mathbb{Z}^n$, and for $1 < j \leq k$

$$f_j = a_j X^{u_j} \prod_{i=1}^{j-1} f_i^{r_{ji}} + b_j X^{v_j} \prod_{i=1}^{j-1} f_i^{s_{ji}},$$

with $a_j, b_j \in \mathbb{R}$, $u_j = (u_{j1}, \dots, u_{jn}) \in \mathbb{Z}^n$, $v_j = (v_{j1}, \dots, v_{jn}) \in \mathbb{Z}^n$, $r_{ji} \in \mathbb{Z}$, $s_{ji} \in \mathbb{Z}$ for $1 \leq i < j$, and

$$f = c X^w \prod_{j=1}^k f_j^{t_j}$$

with $c \in \mathbb{R}$, $w = (w_1, \dots, w_n) \in \mathbb{Z}^n$ and $t_1, \dots, t_k \in \mathbb{Z}$. In other words, starting with the variables $X_1, \dots, X_n$ and real constants and allowing addition, multiplication and division one can obtain f using at most k additions (and an unlimited number of multiplications and divisions.)

Let us say that a semialgebraic set $A \subseteq \mathbb{R}^n$ has **additive complexity at most** $(n, p, k) \in \mathbb{N}^3$ if

$$(*) \quad \left\{ \begin{array}{l} A = \bigcup_{i \in I} A_i, \text{ where } I \text{ is a finite index set and for each } i \in I \\ A_i = \{x \in \mathbb{R}^n : f_1^i(x) = \dots = f_{q_i}^i(x) = 0, g_1^i(x) > 0, \dots, g_{r_i}^i(x) > 0\}, \\ \text{with polynomials } f_\kappa^i, g_\lambda^i \in \mathbb{R}[X], \text{ all of additive complexity at most } k, \\ \text{and } \sum_{i \in I} (q_i + r_i) \leq p. \end{array} \right.$$

Then we have the following finiteness result conjectured by Benedetti and Risler, see pp. 214–215 of [2].

(3.5) THEOREM. *Given any triple $(n, p, k) \in \mathbb{N}^3$ there are only finitely many different embedded homeomorphism types of semialgebraic subsets of $\mathbb{R}^n$ of additive complexity at most (n, p, k) .*

Note that this generalizes proposition (3.2) above on zero sets of fewnomials. As with that proposition, the theorem we just stated follows from the fact that, given (n, p, k) , there are sets $S_1, \dots, S_M \subseteq \mathbb{R}^{N+n}$ (for suitable M, N depending on (n, p, k)), each definable in $\mathbb{R}_{\text{exp}}$, such that each semialgebraic set $A \subseteq \mathbb{R}^n$ of additive complexity at most (n, p, k) is of the form $A = (S_m)_c$ for some $m \in \{1, \dots, M\}$ and some $c \in \mathbb{R}^N$. To prove this fact we may assume that not only (n, p, k) is given, but also the finite index set I and the natural numbers q_i ($i \in I$) and r_i ($i \in I$) appearing in the representation $(*)$ of the semialgebraic sets A we are considering. The desired result will then follow easily from the next two lemmas by the same kind of argument as used in the proof of (3.2).

(3.6) LEMMA. *A rational function $f \in \mathbb{R}(X)$ has additive complexity $\leq k$ if and only if there is a sequence of nonzero polynomials $g_1, \dots, g_k \in \mathbb{R}[X]$ such that*

(i)

$$g_1 = a_1 X^{\alpha_1} + b_1 X^{\beta_1},$$

$$\text{with } a_1, b_1 \in \mathbb{R}, \alpha_1 = (\alpha_{11}, \dots, \alpha_{1n}) \in \mathbb{N}^n, \beta_1 = (\beta_{11}, \dots, \beta_{1n}) \in \mathbb{N}^n,$$

(ii) for $1 < j \leq k$,

$$g_j = a_j X^{\alpha_j} \prod_{i=1}^{j-1} g_i^{d_{ji}} + b_j X^{\beta_j} \prod_{i=1}^{j-1} g_i^{e_{ji}},$$

$$\text{with } a_j, b_j \in \mathbb{R}, \alpha_j = (\alpha_{j1}, \dots, \alpha_{jn}) \in \mathbb{N}^n, \beta_j = (\beta_{j1}, \dots, \beta_{jn}) \in \mathbb{N}^n, d_{ji} \in \mathbb{N}, e_{ji} \in \mathbb{N} \text{ for } 1 \leq i < j, \text{ and}$$

(iii)

$$f \cdot X^\gamma \prod_{j=1}^k g_j^{r_j} = c X^\delta \prod_{j=1}^k g_j^{s_j}$$

with $c \in \mathbb{R}$, $\gamma = (\gamma_1, \dots, \gamma_n) \in \mathbb{N}^n$, $\delta = (\delta_1, \dots, \delta_n) \in \mathbb{N}^n$, and $r_1, \dots, r_k, s_1, \dots, s_k \in \mathbb{N}$.

PROOF. If such a sequence $g_1, \dots, g_k$ exists, then f is clearly of additive complexity $\leq k$. Suppose f is of additive complexity $\leq k$, and let $f_1, \dots, f_k$ be a sequence of nonzero rational functions as in (3.4). By writing the integer exponents in the expressions in (3.4) as differences of natural numbers and clearing denominators one obtains a sequence $g_1, \dots, g_k$ of nonzero polynomials in $\mathbb{R}[X]$ satisfying (i) and (ii) such that

$$f_j = g_j \left/ X^{\epsilon_j} \prod_{i=1}^{j-1} g_i^{p_{ji}} \right.$$

for $1 \leq j \leq k$, with $\epsilon_j = (\epsilon_{j1}, \dots, \epsilon_{jn}) \in \mathbb{N}^n$ and $p_{ji} \in \mathbb{N}$ for $1 \leq i < j$. Then f has the form described in (iii). $\square$

(3.7) LEMMA. Suppose $f \in \mathbb{R}[X]$ has additive complexity $\leq k$. Let $Y = (Y_1, \dots, Y_k)$ be a tuple of new variables. Then there are polynomials $h_1(X, Y), \dots, h_k(X, Y) \in \mathbb{R}[X, Y]$ such that each h_j has at most three monomials, and there are two monomials $\mu_1(X, Y)$ and $\mu_2(X, Y)$ and a real constant c such that:

- (i) for each $x \in \mathbb{R}^n$ there is a unique $y \in \mathbb{R}^k$ with $h_1(x, y) = \dots = h_k(x, y) = 0$;
- (ii) for each $x \in \mathbb{R}^n$, if $y \in \mathbb{R}^k$ and $h_1(x, y) = \dots = h_k(x, y) = 0$, then $\mu_1(x, y)f(x) = c\mu_2(x, y)$;
- (iii) the set

$$\{x \in \mathbb{R}^n : \text{if } y \in \mathbb{R}^k \text{ and } h_1(x, y) = \dots = h_k(x, y) = 0, \text{ then } \mu_1(x, y) \neq 0\}$$

is open and dense in $\mathbb{R}^n$.

PROOF. Take polynomials $g_1, \dots, g_k \in \mathbb{R}[X]$ as in the previous lemma. Then we put

$$h_1 := Y_1 - (a_1 X^{\alpha_1} + b_1 X^{\beta_1}),$$

$$\text{and for } 1 < j \leq k, \quad h_j := Y_j - \left(a_j X^{\alpha_j} \prod_{i=1}^{j-1} Y_i^{d_{ji}} + b_j X^{\beta_j} \prod_{i=1}^{j-1} Y_i^{e_{ji}} \right).$$

Also, we put $\mu_1(X, Y) := X^\gamma \prod_{j=1}^k Y_j^{r_j}$ and $\mu_2(X, Y) := X^\delta \prod_{j=1}^k Y_j^{s_j}$ with $\gamma, \delta, r_1, \dots, r_k$ and $s_1, \dots, s_k$ as in the previous lemma. Clearly

$$h_1(x, y) = \dots = h_k(x, y) = 0 \Leftrightarrow y_j = g_j(x) \text{ for } j = 1, \dots, k,$$

from which (i) and (ii) follow immediately. Since $g_1, \dots, g_k$ are nonzero polynomials we also get property (iii). $\square$

(3.8) To see how the last lemma is to be applied, note that conditions (i), (ii) and (iii) definably determine the function $x \mapsto f(x): \mathbb{R}^n \rightarrow \mathbb{R}$ in terms of the polynomials $h_1, \dots, h_k$, the monomials μ_1 and μ_2 , and the constant c : it is the unique continuous extension to $\mathbb{R}^n$ of the function $x \mapsto c\mu_2(x, y)/\mu_1(x, y)$ defined on the open dense set in (iii), where $y \in \mathbb{R}^k$ is given by $h_1(x, y) = \dots = h_k(x, y) = 0$.

(3.9) EXERCISE. Show that there is no function $F: \mathbb{R}^2 \rightarrow \mathbb{R}$, definable with parameters in $\mathbb{R}_{\text{exp}}$, such that $F(x, k) = x^k$ for all $x \in \mathbb{R}$ and $k \in \mathbb{N}$.

Hint: use that $\mathbb{R}_{\text{exp}}$ is o-minimal.

Notes and comments

The semialgebraic trivialization theorem over $\mathbb{R}$ is due to Hardt [29], who notes that his arguments go through for the category of bounded subanalytic sets. For similar results about the category of Nash manifolds, see Coste and Shiota [10]. Here we follow closely the treatment of the semialgebraic case over arbitrary real closed fields in Delfs and Knebusch [11]. There is a difficulty similar to the one in the previous chapter. This explains why the key lemma (1.5) was obtained in a somewhat roundabout way (first doing the bounded case, and then reducing to that case) compared to the route available for the corresponding semialgebraic result (6.3) in [11].

Subsequently (in 1993) I found the easy theorem (2.2) of Chapter 6, which leads to the short model-theoretic proof in Chapter 8, (3.13) and (2.14), that triangulation implies trivialization. But it seemed reasonable to keep also the longer but quite explicit “geometric” construction of the trivialization. The local conical structure of definable sets (theorem (2.3)) is a familiar consequence of trivialization, as in the semialgebraic case treated in [4].

As soon as Wilkie’s theorem was available (in 1991) it was clear that one could use the trivialization theorem with distinguished subsets to answer the question of Risler and Benedetti as done in Section 3.

The following came too late to my attention to include in the references at the end of the book. It is closely related to the material of this chapter.

M. Coste, *Topological types of fewnomials*, preprint, December 1996.